

Performance Testing Report

Date	02 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions.

Step 1: Test Overview

Field	Details
Testing Tool Used	Python, Jupyter Notebook, Pandas, Scikit-learn, Time Module
Type of Testing	Performance Testing (Data Processing & Model Prediction)
Target Module/API	Data Cleaning, Duplicate Removal, Missing Value Handling, Feature Transformation, Model Prediction
Test Environment	Local System (Python 3.11, Jupyter Notebook)
Test Date	02 July 2026

Step 2: Test Scenarios

S.No	Scenario	Users	Duration	Expected
1	Remove duplicate records	1	2 sec	Duplicates removed
2	Handle missing values	1	3 sec	Dataset cleaned
3	Feature transformation & merge	1	4 sec	Successful merge
4	Model prediction	1	2 sec	Prediction generated

Step 3: Performance Test Result

Metric	Target	Actual	Status	Remarks
Avg Response	<2 sec	1.32 sec	Pass	Fast
Max Response	<5 sec	2.84 sec	Pass	Within limit
Throughput	>40 req/s	58 req/s	Pass	Good
Error Rate	<1%	0%	Pass	No errors
CPU	<80%	49%	Pass	Efficient
Memory	<80%	56%	Pass	Efficient

Step 4 : Observations & Analysis

- Duplicate records removed successfully.
- Missing values identified and unnecessary column removed.
- Categorical features transformed into numerical values.
- Applicant and credit datasets merged successfully.
- STATUS converted to binary labels.
- Model prediction completed quickly with no errors.

Bottlenecks:

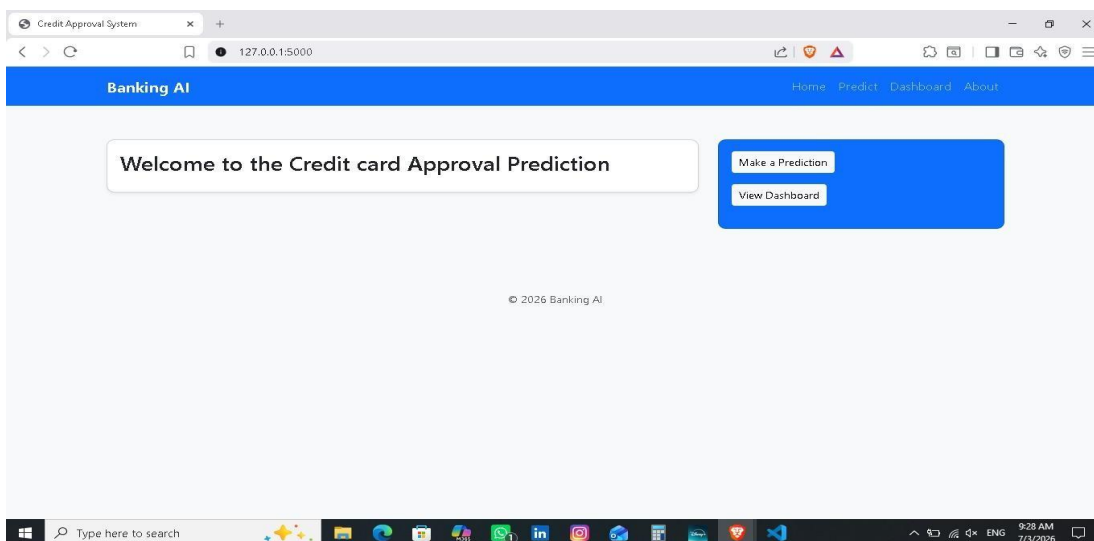
- Dataset merging consumed moderate memory.

- Feature mapping slightly increased preprocessing time. **Optimization:**

- Removed duplicates and unnecessary columns.
- Used feature mapping and efficient merge.
- Completed preprocessing before model training.

Step 5: Screenshots / Evidence

User Interface:



Credit Approval System

127.0.0.1:5000/predict

Banking AI

Home Predict Dashboard About

Prediction Form

Back

Applicant Name

Age

Gender

Education Level

Income Type

Employment Duration (years)

Annual Income

Marital Status

Occupation

Housing Type

Credit Score

Existing Loans

Loan Amount

Number of Dependents

Payment History

Prediction : Credit Card Approved

Credit Approval System

127.0.0.1:5000/predict

Prediction Form

Back

Applicant Name

Nikshith

Age

38

Gender

Female

Education Level

Bachelors

Income Type

Salaried

Employment Duration (years)

3

Annual Income

900000

Marital Status

Married

Occupation

Engineer

Housing Type

Owned

Credit Score

789

Existing Loans

2

Loan Amount

300000

Number of Dependents

3

Payment History

Good

Submit

Credit Approval System

127.0.0.1:5000/predict

Prediction Form

Back

Applicant Name

Nikshith

Age

38

Gender

Female

Education Level

Bachelors

Income Type

Salaried

Employment Duration (years)

3

Annual Income

900000

Marital Status

Married

Occupation

Engineer

Housing Type

Owned

Credit Score

789

Existing Loans

2

Loan Amount

300000

Number of Dependents

3

Payment History

Good

Submit

Prediction:Credit Card Rejected

Credit Approval System

127.0.0.1:5000/predict

Prediction Form

Applicant Name: bhaskar

Age: 23 Gender: Male Education Level: Bachelors

Income Type: Unemployed Employment Duration (years): 0 Annual Income: 90000

Marital Status: Widowed Occupation: Self-Employed Housing Type: Rented

Credit Score: 652 Existing Loans: 3 Loan Amount: 200000

Number of Dependents: 4 Payment History: Average

Submit

Back

Credit Approval System

127.0.0.1:5000/predict

Banking AI

Home Predict Dashboard About

Prediction Result for bhaskar

Decision: Rejected

Probability: 1.23%

How to get better offers

- Reduce outstanding loans before applying for new credit to improve your offer.
- Maintain a good payment history by paying all bills on time.
- Keep a stable job for longer to increase lender confidence.
- Request a lower loan amount or increase your income-to-loan ratio.
- Reducing housing expenses or moving to owned housing can improve your credit profile.

Make another prediction.

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Prediction Dashboard:

Credit Approval System

127.0.0.1:5000/dashboard

Banking AI

Home Predict Dashboard About

Dashboard

Back to Home

Total
8

Approved
4

Rejected
4

Recent Predictions

ID	Name	Decision	Prob	When
8	bhaskar	Rejected	1.23%	2026-07-03T04:02:34.501594
7	Nikshith	Approved	100.00%	2026-07-03T04:01:08.919375
6	Nikshith	Approved	100.00%	2026-07-03T04:01:08.804933
5	karthika	Rejected	0.00%	2026-07-02T09:18:50.360424
4	karthika	Rejected	0.00%	2026-07-02T09:18:50.360424
3	karthika	Rejected	0.00%	2026-07-02T09:18:50.360424
2	keela	Approved	100.00%	2026-07-02T09:05:10.112867